

Khalifeh Basiri

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EDUCATION

Carleton University

Bachelor of Computer Science, GPA: 3.5/4.0

Ottawa, ON

2026

WORK EXPERIENCE

Fisheries and Oceans Canada (DFO) | Government of Canada

Software Developer (internship)

Ottawa, ON

Aug. 2024 – Dec. 2025

- Built a bilingual Python geographic-name validator using the GNBC REST API and geospatial matching, replacing manual entry-by-entry file checks with validation completed in minutes.
- Expanded the validator into a modular Name Manager with comparison, GNBC search, and geospatial helper workflows; CHS Atlantic reported one process fell from two weeks to a few hours.
- Led a three-person Agile team and worked directly with DFO clients through requirements, design, development, demos, deployment, and user training for the Name Manager.
- Tested API, database, multithreading, and geospatial workflows through integration, regression, and user-acceptance testing (UAT); validated releases with users across five DFO regions and the Canadian Coast Guard and supported the Name Manager in production.

Programmer (internship)

Sep. 2023 – Aug. 2024

- Engineered a multithreaded Python Storage Explorer using Azure Blob Inventory, MongoDB, and SQLite to search and export metadata across a 420 TB+ Azure storage estate; cut DFO staff's file-location searches from hours to seconds or minutes.
- Implemented metadata refresh handling that kept searches responsive and preserved the last valid database after failed updates, allowing staff to continue using the Explorer.

Floxy

Frontend Developer

Remote

Mar. 2025 – May 2025

- Built Floxy's responsive Next.js site with React, TypeScript, and Tailwind CSS for a platform with 20,000+ customers; the site reached 23.9K monthly visits within four months.
- Organized the site into reusable components and centralized content models, keeping product pages and language-specific API examples consistent and easy to update.

PROJECTS

ZDash | TypeScript, React, Electron, SQLite, OpenAI API, Zod

[Product Site](#)

- Engineered a Windows diagnostics and ownership app for Nissan and Infiniti CONSULT-II ECUs with live telemetry, safety-gated fault-code workflows, and SQLite/WAL session recording.
- Integrated an AI assistant with vehicle readings and service history through validated, user-approved actions; prototyped RAG over 8,618 service-manual pages with PostgreSQL/pgvector hybrid search and page-level citations.

Tandem t:slim X2 Insulin Pump Simulator | C++, Qt, SQLite, UML

[Project](#)

- Designed bolus calculations and CGM-driven insulin-control logic in a four-person C++/Qt team, adding safety limits, suspend/resume behavior, alerts, and SQLite-backed event history.

Real-Time Sign Language Translator | Machine Learning, Python, TensorFlow Lite

[GitHub](#) | [Demo](#)

- Developed a machine-learning-powered ASL alphabet recognizer; transformed 21 MediaPipe hand landmarks into 63-feature vectors for confidence-scored TensorFlow Lite predictions through a Flask API and React webcam interface.

Collaborative Board | Next.js, TypeScript, Convex, Clerk

[GitHub](#) | [Demo](#)

- Built and deployed a real-time community platform with Next.js, Convex, and Clerk, implementing reactive updates, server-side authorization, threaded comments, per-user vote state, ownership checks, and cascade deletion.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C++, C, SQL, HTML, CSS

Frameworks and libraries: React, Next.js, Node.js, Electron, Flask, Qt, Qt Charts, TensorFlow, TensorFlow Lite, MediaPipe, Tailwind CSS

Cloud, APIs, and tools: Microsoft Azure, Azure Blob Storage, QGIS, Docker, Git, Linux, REST APIs, OpenAI API

Databases: MongoDB, SQLite, Oracle, PostgreSQL, Convex